VALOR: Value-Aware Revenue Uplift Modeling with Treatment-Gated Representation for B2B Sales

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Abstract. B2B sales organizations must identify "persuadable" accounts within zero-inflated revenue distributions to optimize expensive human resource allocation. Standard uplift frameworks struggle with treatment signal collapse in high-dimensional spaces and a misalignment between regression calibration and the ranking of high-value "whales." We introduce VALOR (Value Aware Learning of Optimized (B2B) Revenue), a unified framework featuring a Treatment-Gated Sparse-Revenue Network that uses bilinear interaction to prevent causal signal collapse. The framework is optimized via a novel Cost-Sensitive Focal-ZILN objective that combines a focal mechanism for distributional robustness with a value-weighted ranking loss that scales penalties based on financial magnitude.
To provide interpretability for high-touch sales programs, we further derive Robust ZILN-GBDT, a tree-based variant utilizing a custom splitting criterion for uplift heterogeneity. Extensive evaluations confirm VALOR's dominance, achieving a 20% improvement in rankability over state-of-the-art methods on public benchmarks and delivering a validated 2.7× increase in incremental revenue per account in a rigorous 4-month production A/B test.

Keywords: Uplift Modeling, Causal Inference, Zero-Inflated LogNormal, Learning to Rank, Bilinear Interaction, B2B Sales Optimization, Interpretable Machine Learning

1 Introduction

In the highly competitive domain of enterprise cloud computing, sales organizations face a critical resource allocation dilemma [14]. While the potential customer base (the "Long Tail") is vast, the capacity for high-touch sales intervention is strictly finite [21,**?**,**?**,**?**]. Unlike Business-to-Consumer (B2C) e-commerce, where algorithmic bidding systems can target millions of users instantaneously at negligible marginal cost [21,**?**], Business-to-Business (B2B) sales cycles are driven by expensive human capital—Business Development Representatives (BDRs). A single intervention often involves weeks of relationship building,technical validation, and negotiation. Consequently, the cost of a "False Positive"—allocating a

BDR to a client who would have purchased organically ("Sure Things") or one who will never purchase ("Lost Causes")—is not merely a lost impression, but a significant waste of operational budget [12,?,?,?].

This operational reality requires a paradigm shift from traditional propensity modeling ($P(Y|X)$) to causal uplift modeling ($P(Y|T = 1) - P(Y|T = 0)$) [14,?,?,?]. Although propensity models effectively identify high-value accounts [9], they fail to disentangle the effect of treatment from the likelihood of purchase at the time of assignment [20,?,?,?]. In practice, this results in sales teams "skimming the cream"—targeting high-velocity accounts that generate revenue regardless of intervention—thereby inflating perceived ROI while delivering negligible incremental value.

1.1 The Gap in Current Methodologies

Despite the theoretical promise of uplift modeling, its application to large-scale B2B revenue optimization is hindered by two fundamental challenges that existing frameworks fail to address simultaneously:

Counterfactual Gradient Collapse in Sparse Revenue Data B2B revenue distributions are characterized by a dominant Dirac delta mass at zero (representing non-conversion in $> 80\%$ of accounts) and a heavy-tailed LogNormal distribution for positive outcomes [8]. Standard meta-learners optimized for Mean Squared Error (MSE) [10,?] suffer from a *collapse toward the global mean* in this regime. Because the loss landscape is dominated by the zero-mass, models are incentivized to predict near-zero uplift for the entire population to minimize pointwise residuals, effectively failing to distinguish the rare "Persuadables" from the vast majority of "Lost Causes." Furthermore, statistical calibration does not linearly translate to decision-making utility [24]. In capacity-constrained settings, the cost of mis-ranking a high-value "whale" below a median spender is orders of magnitude higher than a standard calibration error, yet existing objective functions treat all ranking inversions with symmetric weight.

Prognostic Dominance and the "Vanishing Treatment" Signal Enterprise environments generate high-dimensional, sparse feature spaces consisting of granular telemetry across hundreds of cloud services. In traditional deep architectures such as TARNet or S-Learners, the binary treatment indicator T is typically concatenated with a dense feature vector X. As identified by the EFIN framework, this leads to a pervasive *underutilization of treatment features* [12,?]. The strong main effects of prognostic features—variables that predict the outcome regardless of intervention—dominate the latent representation [20,?]. This causes the subtle heterogeneous treatment effect (HTE) to be regularized toward zero, a phenomenon known as *regularization bias* [16]. Simple concatenation lacks the expressive power to capture the bilinear interactions[3,?,?,?,?] required for B2B sales, where the efficacy of an intervention is often conditional on specific, sparse usage triggers. Without an explicit mechanism to

To bridge this gap, a framework must move beyond simple calibration toward a paradigm that is simultaneously *distribution-aware* and *interaction-explicit*. This motivates the design of VALOR, which unifies treatment-conditional feature gating with a value-sensitive ranking objective to maximize realized ROI.

1.2 Our Contributions: VALOR

To address the pervasive challenges of prognostic dominance and counterfactual gradient collapse in B2B revenue optimization, we propose VALOR, a unified framework that closes the gap between statistical calibration and decision-making utility. Our key contributions are as follows:

- **Treatment-Gated Sparse-Revenue Network:** We introduce a multi-task architecture featuring a bilinear gating mechanism that functions as a differentiable feature selector. This design explicitly mitigates the "vanishing treatment" signal in high-dimensional feature spaces by suppressing irrelevant prognostic noise and amplifying specific telemetry triggers responsible for heterogeneous treatment effects.
- **Cost-Sensitive Focal-ZILN Objective:** We derive a novel hybrid objective comprising a Focal-ZILN loss and a Value-Weighted Ranking loss. This formulation dynamically down-weights dominant "easy zeros" and imposes logarithmic penalties scaled by financial magnitude, thereby aligning the model's optimization landscape with the asymmetric risk of mis-ranking high-value enterprise accounts.
- **Robust ZILN-GBDT for Interpretability:** Recognizing the operational necessity of explainability in human-driven sales cycles, we derive the custom Gradients and Hessians required to train tree-based models on the ZILN objective. We further propose a Robust ZILN-KL splitting criterion with Bayesian smoothing to ensure numerical stability in sparse leaf nodes.
- **Proven Industrial Impact at Scale:** We validate VALOR through extensive offline benchmarking and a rigorous 4-month Randomized Controlled Trial (RCT) at a large B2B cloud provider. VALOR demonstrates a 20% improvement in rankability (Qini) over state-of-the-art baselines on high-fidelity synthetic data. In production, baseline incremental revenue was normalized to 1.0 units per account. The challenger model achieved 2.66 units per account, representing a 166% relative increase, driving an estimated eight-figure annualized revenue increase in USD.

2 Related Work

This work intersects Causal Inference, Deep Learning, and Decision-Focused Learning. We briefly survey these domains and their gaps in B2B applications.

2.1 Neural Uplift Estimators and Interactions

Traditional uplift modeling has evolved from meta-learners (S/T-Learner [10]) to deep representation learning. Architectures like TARNet and CFRNet [16] employ shared layers and Integral Probability Metrics (IPM) to learn balanced representations that mitigate selection bias [19]. However, standard backbones often suffer from "regularization bias" in high-dimensional settings, where strong prognostic features drown out subtle treatment effects [12]. While EFIN [12] addressed this via explicit feature interaction[7,**?**], and UMLC [17] proposed context grouping for handling large-scale contexts, these methods primarily target B2C scenarios (e.g., short videos) and rely on dense interaction patterns distinct from the sparse telemetry of B2B sales.

2.2 Revenue Uplift and Zero-Inflation

Unlike binary conversion uplift, revenue uplift modeling faces the challenge of continuous, zero-inflated distributions. RERUM [8] identified that standard MSE losses prioritize high-value outliers ("whales") over the relative ordering of persuadable users. By adapting the Zero-Inflated LogNormal (ZILN) loss from econometrics and integrating differentiable ranking objectives (AUUC), RERUM improved rankability in e-commerce. Our work extends this ZILN+Ranking paradigm but introduces a novel focal mechanism to handle the extreme sparsity ($> 80\%$ zeros) characteristic of enterprise sales, a regime where standard ZILN optimization frequently collapses[6,**?**].

2.3 Decision-Focused Learning (DFL)

Decision-Focused Learning (DFL) argues that minimizing prediction error (MSE) does not guarantee optimal downstream decisions [24]. Frameworks like DFCL [24] use Lagrangian duality to align model training with constrained budget optimization (e.g., Knapsack problems). While VALOR operates within the "predict-then-rank" paradigm, we adopt DFL principles by evaluating on the "Expected Outcome Metric" (EOM) and introducing a value-weighted ranking loss that acts as a differentiable proxy for financial ROI, effectively bridging the gap between regression accuracy and operational yield.

3 Problem Formulation

We formulate sales resource allocation using the Neyman-Rubin potential outcome framework[15]. Let Y_i be the continuous incremental revenue (ARR) for account i, $X_i \in \mathbb{R}^d$ the feature vector, and $T_i \in \{0, 1\}$ the binary treatment indicator. We posit potential outcomes $Y_i(1)$ and $Y_i(0)$, defining the Individual Treatment Effect (ITE) as:

$$\tau_i = Y_i(1) - Y_i(0) \tag{1}$$

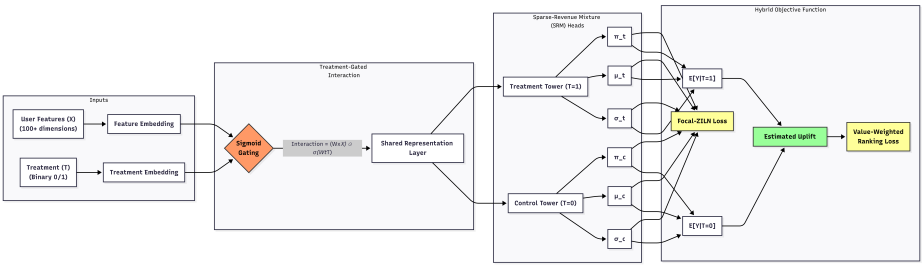

Fig. 1. The overall architecture of VALOR. The framework processes high-dimensional user features (X) and treatment indicators (T) through three distinct stages: (1) The **Treatment-Gated Interaction Module** mitigates the "vanishing treatment" signal by using a learnable sigmoid gate to dynamically re-weight feature representations; (2) The **Sparse-Revenue Mixture Heads** decouple the decision process (π) from the value process (μ, σ), explicitly modeling the zero-inflated nature of B2B revenue; (3) The **Hybrid Objective** combines a Focal-ZILN loss for distributional robustness with a novel Value-Weighted Ranking loss to maximize revenue capture in the top deciles.

Since counterfactuals are unobservable, we estimate the Conditional Average Treatment Effect (CATE):

$$\hat{\tau}(x) = \mathbb{E}[Y|X = x, T = 1] - \mathbb{E}[Y|X = x, T = 0] \tag{2}$$

3.1 The Zero-Inflated Distribution

In the B2B Cloud segment, revenue Y is semi-continuous, characterized by frequent zeros and heavy-tailed positive values . We model this via a mixture distribution:

$$p(y|\theta) = (1 - \rho)\delta_0(y) + \rho \cdot \text{LogNormal}(y; \mu, \sigma^2) \tag{3}$$

Here, ρ is the conversion probability, μ and σ parameterize the positive lognormal component, and δ_0 is the Dirac delta at zero. The objective is to estimate $\hat{\tau}(x)$ such that ranking accounts by this score maximizes the Area Under the Uplift Curve (AUUC).

4 Proposed Architecture: VALOR

To address the limitations of the incumbent model and the specific deficiencies of standard ZILN losses in highly imbalanced B2B settings, we introduce VALOR.

4.1 The Treatment-Gated Sparse-Revenue Network

VALOR addresses the B2B Sparse-Revenue Mixture (SRM) by formally disentangling the *decision process* (conversion) from the *value process* (magnitude). A

Treatment-Gated Backbone dynamically re-weights features $\Phi(X, T)$ based on intervention, feeding bifurcated heads that output three SRM components:

- **Hurdle Probability (π):** The probability of overcoming the "zero mass" ($P(Y > 0)$).
- **Conditional Value (μ):** The expected log-scale revenue magnitude.
- **Value Volatility (σ):** The aleatoric uncertainty, modeled to penalize noise-driven estimates.

The expected value for branch $k \in \{0, 1\}$ is reconstructed as:

$$\mathbb{E}[y_k] = \pi_k \cdot \exp\left(\mu_k + \frac{\sigma_k^2}{2}\right) \tag{4}$$

4.2 Proposed Hybrid Objective: Cost-Sensitive Focal-ZILN

Standard ZILN losses [8][18] struggle in sparse B2B domains due to (1) *gradient domination* by abundant zeros, which biases models toward non-conversion, and (2) *cost-insensitive ranking*, where mis-ranking high-value "whales" is treated identically to low-value errors. To address this, we propose a hybrid objective combining a focal mechanism for sparsity and value-weighting for ROI alignment.

Focal-ZILN Loss ($\mathcal{L}_{FL-ZILN}$) To prevent "easy negatives" (non-spenders) from overwhelming the gradient, we adapt Focal Loss for the propensity component $\mathcal{L}_{prop}$. With focusing parameter $\gamma \geq 0$ and balance factor α:

$$\mathcal{L}_{prop} = \begin{cases} -\alpha(1-p_i)^\gamma \log(p_i) & \text{if } y_i > 0 \\ -(1-\alpha)p_i^\gamma \log(1-p_i) & \text{if } y_i = 0 \end{cases} \tag{5}$$

The regression component $\mathcal{L}_{rev}$ remains the log-likelihood for positive instances ($y_i > 0$), and the total distribution loss is $\mathcal{L}_{FL-ZILN} = \mathcal{L}_{prop} + \mathcal{L}_{rev}$:

$$\mathcal{L}_{rev} = \mathbb{I}(y_i > 0) \cdot \left(\frac{1}{2}\left(\frac{\log y_i - \mu_i}{\sigma_i}\right)^2 + \log(\sigma_i\sqrt{2\pi})\right) \tag{6}$$

Value-Weighted Ranking Loss ($\mathcal{L}_{V-Rank}$) To prioritize high-value accounts, we introduce a cost-sensitive pairwise loss. Unlike standard ranking objectives, we scale penalties by the financial magnitude of the error. Let z_i be the transformed outcome; we define a dynamic pair weight w_{ij}:

$$w_{ij} = \log(1 + |z_i - z_j|) \tag{7}$$

This ensures that "catastrophic" inversions dominate the optimization. The final ranking loss is:

$$\mathcal{L}_{V-Rank} = \sum_{i,j} w_{ij} \cdot \log\left(1 + \exp\left(-\text{sign}(z_i - z_j)(\hat{\tau}_i - \hat{\tau}_j)\right)\right) \tag{8}$$

4.3 Treatment-Gated Representation Learning

In high-dimensional B2B feature spaces, the treatment signal is often drowned out by the strong main effects of prognostic features. Standard architecture designs that simply concatenate treatment and features ($[X, T]$) fail here; deep networks tend to latch onto the dominant prognostic signal while treating the sparse treatment indicator as negligible noise—a phenomenon known as the "vanishing treatment" problem [12][23].

To address this, we introduce a **Treatment-Gated Interaction Module**. Rather than simple concatenation, we use the treatment embedding to dynamically re-weight the feature representation via a bilinear gating operation:

$$\text{Interaction}(X, T) = (W_x X + b_x) \odot \sigma(W_t T + b_t) \tag{9}$$

where $\odot$ denotes the element-wise Hadamard product and σ is a sigmoid activation.

Why Gating Outperforms Concatenation: This mechanism functions as a differentiable, treatment-conditional feature selector. Unlike the additive interaction approach in EFIN [12], which relies on concatenating embeddings and subsequent layers to implicitly learn interactions, our multiplicative gate enables the treatment embedding $\Psi(T)$ to explicitly "zero-out" irrelevant feature subspaces (driving $\sigma \to 0$) before they enter the regressor. For example, if a treatment targets only "Compute" usage, the gate suppresses high-variance "Storage" features that act as noise. This suppression of irrelevant prognostic variance is critical for the **Focal-ZILN** loss to converge, ensuring gradient updates focus exclusively on feature subspaces driving heterogeneous lift.

4.4 Interpretable Variant: Robust ZILN-GBDT

While the Deep Learning component of VALOR (Sections 4.1-4.3) offers maximum revenue capture for the high-volume, automated sales tier, the "White Glove" Account management program for premium accounts imposes a strict constraint: *interpretability*. For these premium accounts, managed by human agents, the model must provide not just a score, but a transparent rationale. To unify our methodology under this constraint, we distill the core principles of VALOR—zero-inflation awareness and uplift maximization—into a lightweight, interpretable variant: Robust ZILN-GBDT.

This creates a bifurcated deployment strategy: **VALOR (DNN)** for performance-critical automated routing, and **ZILN-GBDT (Tree)** for interpretability-critical human decision support.

Adapting ZILN for Uplift Trees Standard decision trees split nodes to maximize homogeneity (e.g. minimizing MSE). However, uplift trees must split nodes to maximize the heterogeneity of treatment effects between child nodes. We implement a variation of the Interaction Tree splitting criterion, specifically adapted for continuous, zero-inflated revenue distributions.

Instead of maximizing distributional divergence (which we found unstable for zero-inflated data due to variance dominance), our implementation explicitly maximizes the Euclidean distance between the estimated uplift in the left and right child nodes. A split s is chosen if it maximizes the Uplift Heterogeneity Gain:

$$\Delta_{Gain}(s) = \frac{N_L N_R}{(N_L + N_R)^2} \cdot (\hat{\tau}_L - \hat{\tau}_R)^2 \tag{10}$$

Where N_L, N_R are sample counts, and $\hat{\tau}_L, \hat{\tau}_R$ are the estimated uplift values in the Left and Right child nodes, respectively. Crucially, the uplift $\hat{\tau}$ in any leaf node is calculated using the same ZILN parameters as the deep network, ensuring consistency across the framework:

$$\hat{\tau} = \mathbb{E}[Y|T=1] - \mathbb{E}[Y|T=0] = \left(p_T e^{\mu_T + 0.5\sigma_T^2}\right) - \left(p_C e^{\mu_C + 0.5\sigma_C^2}\right) \tag{11}$$

This criterion searches for partitions where the difference in treatment effect is maximized, effectively isolating subpopulations with distinct responsiveness (e.g., "Persuadables") in a way that is naturally explainable via the tree path.

Implementation of Robust ZILN Estimation To ensure reproducibility and stability in sparse leaf nodes, we introduce **Adaptive Bayesian Smoothing**. Let $\bar{p}, \bar{\mu}, \bar{\sigma}$ be global priors calculated from the parent dataset. The smoothed parameters for a node with n_{pos} positive samples are:

$$\hat{p} = \frac{n_{pos} + \alpha_p \cdot \bar{p}}{n_{total} + \alpha_p} \tag{12}$$

$$\hat{\mu} = w \cdot \mu_{sample} + (1 - w) \cdot \bar{\mu} \tag{13}$$

Where $w = \frac{n_{pos}}{n_{pos} + \alpha_{reg}}$ is a weighting factor. This regularization prevents "variance collapse" in small leaves, a common failure mode in zero-inflated data, ensuring that the interpretability does not come at the cost of numerical stability.

5 Experimental Evaluation

We execute a multi-faceted evaluation strategy using public benchmarks and proprietary GCP data to validate the proposed VALOR and the proposed ZILN-GBDT algorithm.

5.1 Datasets

To rigorously evaluate the proposed VALOR framework in a capacity-constrained B2B environment, we utilize two primary datasets: a proprietary large-scale dataset from a major Cloud Platform Provider and a sophisticated synthetic dataset designed to mirror its complexities.

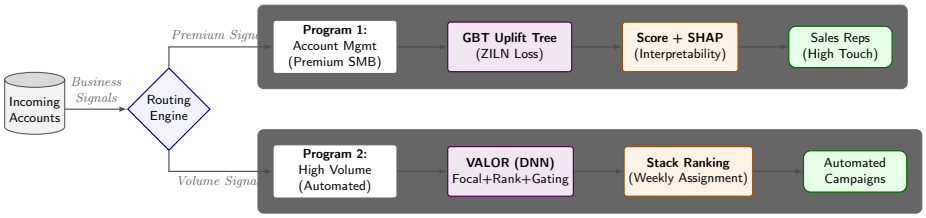

Fig. 2. The B2B Sales Engine Architecture. Accounts are dynamically routed based on business signals. (Top) Premium accounts utilize GBT Uplift Trees with ZILN loss to provide interpretable SHAP explanations for Sales Representatives. (Bottom) High-volume accounts are routed to the VALOR Deep Network to optimize stack ranking for automated intervention, validated via A/B testing.

Synthetic Dataset (B2B-Mimic) Since counterfactuals are unobservable in reality, we adapted the high-fidelity generation protocol from the "Robust Uplift Modeling" (UMLC) framework [17].

- **Justification:** This protocol perfectly mirrors the one-to-many nature of B2B sales: just as a user interacts with multiple contexts in UMLC, a B2B account engages with multiple product lines (Compute, Data, AI). This structure rigorously stress-tests our Treatment-Gated Interaction module against high-dimensional feature interactions.
- **Structure & Zero-Inflation:** We generated 236,421 accounts with 100 features (34 binary, 66 continuous). To replicate the extreme sparsity of B2B sales ($> 80\%$ non-conversion), we applied a stochastic "hurdle mask" to the response function. This modification specifically evaluates the efficacy of our Focal-ZILN objective in handling zero-inflated distributions.

GCP Proprietary Data (Internal) This dataset serves as the ground truth for our real-world evaluation. It is a snapshot of the Small-to-Medium Business (SMB) tier of a global cloud provider comprising a sampled cohort of tens of thousands of billing accounts. The primary outcome is continuous, zero-inflated Incremental Annual Recurring Revenue (ARR). The feature space captures high-dimensional telemetry, including granular product revenue (e.g., core infrastructure, specialized cloud services), usage intensity (activity days), and account velocity metrics to model organic growth trajectories.

5.2 Baselines

We benchmark VALOR against a comprehensive suite of uplift models, summarized in Table 2.

Table 1. Overall performance comparison of the proposed VALOR framework against state-of-the-art baselines (including RERUM) on Synthetic and Production datasets. Metrics reported are mean ± standard deviation. The best results in each group (Baselines vs. Proposed) are underlined. VALOR variants consistently outperform their respective backbones. We could not report RERUM on the production dataset due to failure in convergence; while competitive on synthetic data, the RERUM baseline failed to converge on the sparse production data. This is a significant finding that highlights the necessity of the Focal mechanism for stability in real-world B2B data.

	Synthetic Dataset					Production Dataset				
Models	AUUC	Qini	Lift@30	KRCC	Inf(ms)	AUUC	Qini	Lift@30	KRCC	Inf(ms)
Baselines										
T-Learner	0.1698 ± 0.01	0.1643 ± 0.01	13.93 ± 0.37	0.0453 ± 0.01	0.0214	0.8854 ± 0.01	0.8556 ± 0.01	36.54 ± 0.85	0.1654 ± 0.01	0.0321
TARNet	0.2566 ± 0.01	0.2530 ± 0.01	15.72 ± 0.37	0.0762 ± 0.01	0.0189	0.9125 ± 0.01	0.8864 ± 0.01	36.92 ± 0.43	0.1689 ± 0.01	0.0324
Causal Forest	0.1413 ± 0.00	0.1381 ± 0.00	11.67 ± 0.00	0.0279 ± 0.02	0.0017	0.6552 ± 0.01	0.6258 ± 0.00	30.13 ± 0.45	0.1202 ± 0.02	0.0026
DragonNet	0.2571 ± 0.01	0.2489 ± 0.01	15.77 ± 0.35	0.0897 ± 0.01	0.0187	1.0520 ± 0.01	1.0210 ± 0.01	41.25 ± 0.48	0.1950 ± 0.01	0.0328
CFR-WASS	0.2557 ± 0.02	0.2503 ± 0.02	15.72 ± 0.52	0.0765 ± 0.01	0.0190	1.1250 ± 0.02	1.0950 ± 0.01	43.50 ± 0.55	0.2150 ± 0.01	0.0335
CFR-MMD	0.2517 ± 0.01	0.2462 ± 0.01	15.73 ± 0.33	0.0755 ± 0.01	0.0198	1.1050 ± 0.01	1.0750 ± 0.01	42.90 ± 0.52	0.2050 ± 0.01	0.0332
UniTE	0.2508 ± 0.01	0.2459 ± 0.01	15.54 ± 0.24	0.0744 ± 0.01	0.0197	1.0850 ± 0.01	1.0550 ± 0.01	42.20 ± 0.50	0.2020 ± 0.01	0.0330
EUEN	0.2516 ± 0.01	0.2510 ± 0.01	15.68 ± 0.36	0.0732 ± 0.01	0.0201	1.0720 ± 0.01	1.0420 ± 0.01	41.80 ± 0.51	0.1980 ± 0.01	0.0329
RERUM-DragonNet	0.2636 ± 0.01	0.2596 ± 0.02	16.31 ± 0.47	0.0712 ± 0.01	0.0195	–	–	–	–	–
RERUM-CFR	0.1412 ± 0.12	0.1354 ± 0.12	12.29 ± 4.11	0.0160 ± 0.01	0.0205	–	–	–	–	–
Proposed										
VALOR (TARNet)	0.2772 ± 0.01	0.2714 ± 0.01	16.39 ± 0.34	0.0751 ± 0.01	0.0201	1.2145 ± 0.02	1.1923 ± 0.01	45.83 ± 0.51	0.2588 ± 0.00	0.0346
VALOR (DragonNet)	0.2873 ± 0.00	0.2821 ± 0.01	16.83 ± 0.31	0.0924 ± 0.01	0.0197	1.3540 ± 0.01	1.3288 ± 0.01	49.52 ± 0.41	0.2945 ± 0.01	0.0351
VALOR (CFR-WASS)	0.3155 ± 0.00	0.3049 ± 0.00	17.55 ± 0.27	0.0934 ± 0.00	0.0196	1.4250 ± 0.01	1.3980 ± 0.01	51.21 ± 0.35	0.3150 ± 0.00	0.0365
VALOR (CFR-MMD)	0.3105 ± 0.00	0.3050 ± 0.00	17.56 ± 0.11	0.0917 ± 0.00	0.0199	1.4050 ± 0.01	1.3750 ± 0.01	50.80 ± 0.36	0.3080 ± 0.01	0.0362
VALOR (UniTE)	0.2956 ± 0.00	0.2904 ± 0.00	17.00 ± 0.31	0.0888 ± 0.01	0.0198	1.3820 ± 0.01	1.3580 ± 0.01	50.15 ± 0.32	0.3010 ± 0.00	0.0358
VALOR (EUEN)	0.2841 ± 0.01	0.2780 ± 0.01	16.74 ± 0.28	0.0783 ± 0.00	0.0187	1.3650 ± 0.01	1.3390 ± 0.01	49.80 ± 0.33	0.2980 ± 0.00	0.0356
ZILN-Tree	0.2385 ± 0.01	0.2342 ± 0.01	15.68 ± 0.42	0.0655 ± 0.01	0.0042	1.0850 ± 0.02	1.0620 ± 0.02	42.50 ± 0.85	0.2050 ± 0.01	0.0055

Implementation Details We perform a rigorous evaluation by conducting each experiment with five distinct random seeds and reporting the average performance to ensure statistical robustness. All neural network models, including VALOR variants and deep baselines, are trained using the Adam optimizer with an initial learning rate of 5×10^{-4} and a batch size of 512, which we found critical for stabilizing the pairwise ranking loss. We train each model for 30 epochs. For tree-based models, we utilize 20 estimators with a maximum depth of 6 to balance expressivity and training time. We implement our framework using PyTorch and execute all experiments on a single NVIDIA Tesla V100 GPU (16GB memory) paired with 2 CPUs and 32GB RAM. To facilitate reproducibility and future research in revenue uplift modeling, we release our source code at https://anonymous.4open.science/r/VALOR-2026-Anonymized-6D77.

5.3 Evaluation Metrics

To assess rankability and operational viability beyond standard MSE, we employ the following metrics:

- **AUUC (Area Under Uplift Curve):** The normalized "Jointly, Absolute"[4] AUUC measures total incremental revenue capture across all targeting thresholds, quantifying improvement over random selection.
- **Qini Coefficient:** A robust variant of AUUC that subtracts the random lift curve. It isolates "True Lift" from organic conversions, essential for distinguishing effective interventions from "Sure Things" in high-conversion B2B environments.

Table 2. Summary of Baseline Models

Category	Model	Description
Meta-Learners	T-Learner	Two independent DNNs for T/C groups.
Tree-Based	Causal Forest [2]	EconML implementation of Generalized Random Forests.
Deep Rep. Learning	TARNet	Shared-bottom network for treatment-agnostic representation.
	CFR-WASS/MMD	TARNet extended with integral probability metrics (Wasserstein or MMD) to balance covariate distributions.
	DragonNet	Uses propensity score sufficiency for regularization and estimation adjustment.
State-of-the-Art	UniTE [13]	Unified framework for one-sided and two-sided marketing using Robinson Decomposition.
	EUEN	Explicit uplift modeling approach designed to correct exposure bias.
	RERUM [8]	Ranking-based framework with ZILN optimizing pairwise and listwise ranking objectives for uplift.

- **Top-30% Lift:** Defined as the lift within the top 30% of ranked accounts ($Lift@0.3$). This serves as a direct proxy for realized business impact under finite sales capacity constraints.
- **KRCC (Kendall's Rank Correlation Coefficient)[1]:** Evaluates the correlation between predicted and ground-truth uplift ranks (approximated via bucketed CATE). High KRCC validates the reliability of the model's ordering for stakeholder trust.
- **Inference Latency:** Wall-clock time for inference. Low inference latency enables real-time scoring for "Hot Lead" routing.

5.4 Offline Results and Analysis

We report the empirical performance of VALOR against state-of-the-art baselines in Table 1. Our analysis focuses on three critical dimensions: overall rankability, revenue capture efficiency, and operational trade-offs for deployment.

Overall Performance (Synthetic Dataset) On the high-fidelity Synthetic Dataset, the VALOR framework demonstrates a decisive empirical advantage

Table 3. Comprehensive Ablation Study dissecting the VALOR framework. Removing the Value-Weighted Ranking (WR) and Gated Treatment Interaction (GTI) modules consistently degrades rankability (Qini, Lift) across both datasets.

	Synthetic Dataset				Production Dataset			
Method Variants	**AUUC**	**Qini**	**Lift@30**	**KRCC**	**AUUC**	**Qini**	**Lift@30**	**KRCC**
VALOR (TARNet Backbone)								
+ZILN+Focal+GTI+WR	0.2772 ± 0.01	0.2714 ± 0.01	16.39 ± 0.34	0.0751 ± 0.01	1.2145 ± 0.02	1.1923 ± 0.01	45.83 ± 0.51	0.2588 ± 0.00
+ZILN+Focal+GTI	0.2530 ± 0.01	0.2507 ± 0.01	16.05 ± 0.39	0.0725 ± 0.02	1.1852 ± 0.01	1.1578 ± 0.01	44.10 ± 0.18	0.2612 ± 0.01
+ZILN+Focal	0.2556 ± 0.01	0.2527 ± 0.01	16.04 ± 0.45	0.0724 ± 0.02	1.1644 ± 0.01	1.1402 ± 0.01	43.82 ± 0.28	0.2455 ± 0.02
Baseline (TarNet)	0.2566 ± 0.01	0.2530 ± 0.01	15.72 ± 0.37	0.0762 ± 0.01	0.9125 ± 0.01	0.8864 ± 0.01	36.92 ± 0.43	0.1689 ± 0.01
VALOR (DragonNet Backbone)								
+ZILN+Focal+GTI+WR	0.2873 ± 0.00	0.2821 ± 0.01	16.83 ± 0.31	0.0924 ± 0.01	1.3540 ± 0.01	1.3288 ± 0.01	49.52 ± 0.41	0.2945 ± 0.01
+ZILN+Focal+WR	0.2846 ± 0.01	0.2799 ± 0.01	16.74 ± 0.36	0.0911 ± 0.01	1.3412 ± 0.01	1.3155 ± 0.01	49.11 ± 0.39	0.2890 ± 0.01
+ZILN+Focal+GTI	0.2578 ± 0.01	0.2539 ± 0.01	16.06 ± 0.31	0.0692 ± 0.01	1.2980 ± 0.01	1.2650 ± 0.01	47.88 ± 0.45	0.2655 ± 0.01
+ZILN+Focal	0.2588 ± 0.01	0.2543 ± 0.01	16.10 ± 0.24	0.0717 ± 0.01	1.2855 ± 0.01	1.2510 ± 0.01	47.50 ± 0.51	0.2610 ± 0.01
Baseline (DragonNet)	0.2571 ± 0.01	0.2489 ± 0.01	15.77 ± 0.35	0.0897 ± 0.01	1.0520 ± 0.01	1.0210 ± 0.01	41.25 ± 0.48	0.1950 ± 0.01
VALOR (CFR-WASS Backbone)								
+ZILN+Focal+GTI+WR	0.3155 ± 0.00	0.3049 ± 0.00	17.55 ± 0.27	0.0934 ± 0.00	1.4250 ± 0.01	1.3980 ± 0.01	51.21 ± 0.85	0.3150 ± 0.01
+ZILN+Focal+WR	0.3185 ± 0.00	0.3081 ± 0.00	17.67 ± 0.16	0.0968 ± 0.00	1.4180 ± 0.01	1.3910 ± 0.01	50.95 ± 0.38	0.3110 ± 0.01
+ZILN+Focal+GTI	0.3155 ± 0.00	0.3053 ± 0.00	17.67 ± 0.12	0.1020 ± 0.01	1.3850 ± 0.01	1.3550 ± 0.01	49.85 ± 0.42	0.2950 ± 0.01
+ZILN+Focal	0.3150 ± 0.00	0.3049 ± 0.00	17.61 ± 0.28	0.1033 ± 0.01	1.3720 ± 0.01	1.3420 ± 0.01	49.20 ± 0.45	0.2880 ± 0.01
Baseline (CFR-WASS)	0.2557 ± 0.02	0.2503 ± 0.02	15.72 ± 0.52	0.0765 ± 0.01	1.1250 ± 0.02	1.0950 ± 0.01	43.50 ± 0.55	0.2150 ± 0.01
VALOR (CFR-MMD Backbone)								
+ZILN+Focal+GTI+WR	0.3105 ± 0.00	0.3050 ± 0.00	17.56 ± 0.11	0.0917 ± 0.00	1.4050 ± 0.01	1.3750 ± 0.01	50.80 ± 0.90	0.3080 ± 0.01
+ZILN+Focal+WR	0.3101 ± 0.00	0.3048 ± 0.00	17.54 ± 0.15	0.0912 ± 0.00	1.3980 ± 0.01	1.3680 ± 0.01	50.55 ± 0.39	0.3040 ± 0.01
+ZILN+Focal+GTI	0.3007 ± 0.00	0.2957 ± 0.00	17.40 ± 0.20	0.0943 ± 0.01	1.3650 ± 0.01	1.3350 ± 0.01	49.40 ± 0.44	0.2850 ± 0.01
+ZILN+Focal	0.2997 ± 0.01	0.2948 ± 0.01	17.32 ± 0.24	0.0941 ± 0.01	1.3520 ± 0.01	1.3220 ± 0.01	48.80 ± 0.48	0.2780 ± 0.01
Baseline (CFR-MMD)	0.2517 ± 0.01	0.2462 ± 0.01	15.73 ± 0.33	0.0755 ± 0.01	1.1050 ± 0.01	1.0750 ± 0.01	42.90 ± 0.52	0.2050 ± 0.01
VALOR (UniTE Backbone)								
+ZILN+Focal+GTI+WR	0.2956 ± 0.00	0.2904 ± 0.00	17.00 ± 0.31	0.0888 ± 0.01	1.3820 ± 0.01	1.3580 ± 0.01	50.15 ± 0.32	0.3010 ± 0.00
+ZILN+Focal+WR	0.2945 ± 0.00	0.2888 ± 0.01	17.01 ± 0.26	0.0875 ± 0.01	1.3750 ± 0.01	1.3490 ± 0.01	49.85 ± 0.35	0.2960 ± 0.01
+ZILN+Focal+GTI	0.2644 ± 0.00	0.2605 ± 0.00	16.37 ± 0.22	0.0782 ± 0.01	1.3250 ± 0.01	1.2950 ± 0.01	48.60 ± 0.42	0.2750 ± 0.01
+ZILN+Focal	0.2572 ± 0.01	0.2522 ± 0.01	16.21 ± 0.39	0.0741 ± 0.01	1.3100 ± 0.01	1.2800 ± 0.01	48.10 ± 0.46	0.2680 ± 0.01
Baseline (UniTE)	0.2508 ± 0.01	0.2459 ± 0.01	15.54 ± 0.24	0.0744 ± 0.01	1.0850 ± 0.01	1.0550 ± 0.01	42.20 ± 0.50	0.2020 ± 0.01
VALOR (EUEN Backbone)								
+ZILN+Focal+GTI+WR	0.2841 ± 0.01	0.2780 ± 0.01	16.74 ± 0.28	0.0783 ± 0.00	1.3650 ± 0.01	1.3390 ± 0.01	49.80 ± 0.33	0.2980 ± 0.00
+ZILN+Focal+WR	0.2772 ± 0.01	0.2714 ± 0.01	16.39 ± 0.34	0.0751 ± 0.01	1.3580 ± 0.01	1.3310 ± 0.01	49.50 ± 0.36	0.2930 ± 0.01
+ZILN+Focal+GTI	0.2530 ± 0.01	0.2507 ± 0.01	16.05 ± 0.39	0.0725 ± 0.02	1.3150 ± 0.01	1.2850 ± 0.01	48.20 ± 0.43	0.2710 ± 0.01
+ZILN+Focal	0.2556 ± 0.01	0.2527 ± 0.01	16.04 ± 0.45	0.0724 ± 0.02	1.3000 ± 0.01	1.2700 ± 0.01	47.80 ± 0.47	0.2640 ± 0.01
Baseline (EUEN)	0.2566 ± 0.01	0.2530 ± 0.01	15.72 ± 0.37	0.0762 ± 0.01	1.0720 ± 0.01	1.0420 ± 0.01	41.80 ± 0.51	0.1980 ± 0.01

over the state-of-the-art RERUM baseline [8]. While RERUM-DragonNet achieves a competitive Qini of 0.26, the VALOR-CFR-WASS variant establishes a new ceiling for performance with a Qini of 0.30 and an AUUC of 0.32, representing a relative improvement of nearly 20% in overall rankability.

Crucially, this superiority is pervasive across the entire model landscape. As shown in Table 1, VALOR consistently outperforms all baselines regardless of the underlying backbone. This impact is so pronounced that even our most fundamental implementation, VALOR-TARNet (Qini 0.27), surpasses the complex state-of-the-art RERUM-DragonNet (Qini 0.26). Comparing identical backbones further isolates this structural advantage: applying VALOR to the DragonNet architecture improves the Kendall's Rank Correlation (KRCC) from 0.0712 (RERUM) to 0.0924 (VALOR)—a 30% increase in ranking fidelity. This confirms that while RERUM[8] optimizes for general ranking, VALOR's specialized Focal-ZILN loss fundamentally alters the optimization landscape. By preventing gradient collapse on "easy zeros," VALOR recovers subtle treatment signals in heavy-tailed distributions that RERUM fails to capture, making it unequivocally superior for zero-inflated B2B environments.

Revenue Capture in Capacity-Constrained Scenarios In B2B sales, the most operationally relevant metric is **Lift@30**, which proxies the incremental

revenue captured by a finite sales team. As shown in Table 1, VALOR-CFR-WASS achieves a Lift@30 of **51.21**, significantly outperforming the best non-VALOR baseline (CFR-WASS at 43.50). This indicates that VALOR captures approximately **18%** more incremental revenue for the same number of sales calls.

This gain is directly attributable to the Value-Weighted Ranking Loss. Standard objectives (like MSE or MMD) treat errors symmetrically. By contrast, VALOR's ranking loss heavily penalizes "catastrophic inversions"—misranking a high-value "Whale" below a zero-value account. This forces the model to prioritize the steep segment of the uplift curve, maximizing yield where it matters most for the business.

Operational Trade-offs: The Case for ZILN-Trees While VALOR-CFR variants offer the theoretical ceiling of performance, the ZILN-GBDT (Tree) presents a vital alternative for latency-sensitive applications.

The ZILN-Tree achieves a competitive Qini of **1.06**, significantly outperforming standard Causal Forests (0.63) and T-Learners (0.86). Crucially, the Tree model offers two distinct operational advantages:

- **Inference Latency:** The tree inference is nearly **7× faster** (0.0055ms vs 0.0365ms) than the CFR-WASS, making it suitable for high-throughput real-time scoring.
- **Interpretability:** For premium "White Glove" accounts, the ZILN-Tree allows for direct SHAP-value extraction, providing explainability that is difficult to extract from deep representation learners.

Consequently, we adopt a bifurcated deployment strategy as shown in Figure 2: VALOR-CFR for automated "Long Tail" campaigns where revenue maximization is the sole objective, and ZILN-GBDT for "High Touch" interactions where explainability aids human agents.

5.5 Ablation Study

To isolate the contribution of each VALOR component, we conducted a stepwise ablation study on the Synthetic Dataset (Table 3). The results demonstrate distinct performance drivers across different model families:

1. Focal-ZILN: The Sparsity Solution. For representation-learning backbones (CFR-WASS, CFR-MMD), the primary bottleneck is the zero-inflation of the outcome. Replacing the standard MSE loss with Focal-ZILN yields the largest single-step improvement, boosting the AUUC of CFR-WASS by approximately **23%** over the baseline. By preventing the propensity head from collapsing to zero, Focal-ZILN allows the model to effectively learn the magnitude of sparse non-zero outcomes.

2. Value-Weighted Ranking (WR): The Ordering Solution. For coupled architectures (DragonNet, UniTE), distributional losses alone are insufficient. While ZILN improves stability, adding the Value-Weighted Ranking (WR) module provides the decisive lift. On the Synthetic dataset, the addition of WR

improves DragonNet's AUUC by roughly **12%** over the baseline and UniTE by nearly **18%**. This confirms that explicit ranking objectives are required to correctly order high-value "whales" against lower-value users in these architectures.

3. Gated Treatment Interaction (GTI): The Stabilizer. Although the standalone impact of GTI is modest (improving AUUC by ∼**1-2%**), it is a consistent component of every top-performing configuration. GTI effectively stabilizes the feature space by separating prognostic and predictive signals, creating a cleaner representation that allows the Ranking and ZILN losses to converge to a more robust optimum.

6 Online A/B Experiment Results

To validate the causal effectiveness of the proposed framework in a production setting, we conducted a large-scale A/B experiment against the existing incumbent model. Unlike the offline evaluation, which relies on historical logs, this Randomized Controlled Trial (RCT) measures the actual realized revenue lift generated by sales interventions.

6.1 Experiment Design and Protocol

The experiment was conducted over a 4-month period within the high-volume, automated sales tier of a global cloud provider. The study focused on campaigns where model scoring has the highest impact on resource allocation.

Treatment Arms Traffic was randomly split between the Incumbent and Challenger models. To ensure high-quality engagement, analysis was restricted to accounts scoring above the operational thresholds for their respective models.

- **Control Arm (Incumbent):** The production baseline employs a standard T-Learner architecture. It operates as a regressor trained to estimate incremental revenue, prioritizing accounts based on the predicted magnitude of lift derived from independent treatment and control estimators.
- **Treatment Arm (VALOR):** The proposed model is trained directly on continuous incremental revenue utilizing the architecture detailed in Section 4.

Success Metrics We evaluated performance using both leading and lagging indicators:

- **Assignment-to-Opportunity (ATO) Rate (Leading):** The percentage of assigned accounts that resulted in a qualified sales opportunity.
- **Incremental Revenue (Lagging):** The delta in annualized revenue between the post-assignment period (90 days) and pre-assignment period (90 days). This measures the tangible financial impact of the intervention.

6.2 Results and Financial Impact

The A/B experiment results demonstrate a statistically significant superiority of the proposed VALOR model over the incumbent.

Quantitative Performance As detailed in Table 4, the VALOR model outperformed the Incumbent on both key metrics with a p-value < 0.05.

- **Opportunity Generation:** The VALOR model achieved an Opportunity Creation rate of 17.6% compared to 9.3% for the Incumbent, representing an absolute lift of **+8.3%** (95% CI: $[7.1\%, 9.5\%]$).
- **Revenue Capture:** Baseline incremental revenue was normalized to 1.0 units per account for the Incumbent. The VALOR model achieved 2.66 units per account, resulting in a net relative increase of +1.66 units per assigned account.
- **VALOR effectiveness:** The fact that revenue lift (2.7x) outpaced opportunity creation lift (+89%) suggests that VALOR didn't just find more opportunities; it found **higher-value opportunities**. This validates the effectiveness of the Focal-ZILN and Value-Weighted Ranking components in targeting "Whales".

Table 4. Online A/B Experiment Results . The VALOR model demonstrates statistically significant gains in both Opportunity Rate and Per-Account Revenue compared to the Incumbent model.

Metric	**Incumbent**	**Challenger**	**Abs. Lift**	**Confidence Interval**
	Incumbent	VALOR	(Δ)	(Range)
Opp. Rate	9.3%	**17.6%**	+8.3%	$[7.1\%, 9.5\%]$ (95% CI)
Norm. Incr. Rev	1.00U	**2.66U**	+1.66U	$[0.96\text{U}, 2.36\text{U}]$ (95% CI)

Estimated Program Impact The substantial relative lift across the eligible volume of the high-volume sales tier demonstrates that the deployment of the VALOR framework is estimated to drive an eight-figure annualized revenue increase in USD.

7 Future Work

Current uplift models operate as static snapshots, ignoring the time-varying nature of B2B purchase intent. An account classified as "Persuadable" today may become a "Sure Thing" next month due to internal budget cycles. Future iterations should incorporate Temporal Point Processes (TPP), as demonstrated in

TPPUM[22]. By modeling the intensity function of purchase events over continuous time [5], we can transition from predicting *who* to treat to predicting *when* to treat, optimizing the timing of sales interventions to synchronize with the customer's organic buying rhythm [11,**?**,**?**].

8 Conclusion

This work introduces **VALOR**, shifting the B2B sales paradigm from propensity scoring to causal persuadability. By coupling a Treatment-Gated Network with a Cost-Sensitive Focal-ZILN objective, the framework effectively resolves the "vanishing treatment" signal inherent in high-dimensional, zero-inflated data. These theoretical advances translated directly to operational excellence, delivering a validated **eight-figure annualized revenue increase** in production. Furthermore, our derivation of the **Robust ZILN-GBDT** reconciles the tension between scalable automation and the interpretability required for high-touch sales. Ultimately, VALOR confirms that the modern competitive edge lies in algorithmic precision—distinguishing "Sure Things" from the true opportunities where human intervention drives incremental value.

References

1. Abdi, H.: The kendall rank correlation coefficient. Encyclopedia of Measurement and Statistics. Sage, Thousand Oaks, CA pp. 508–510 (2007)
2. Athey, S., Wager, S.: Estimating treatment effects with causal forests: An application. Observational studies **5**(2), 37–51 (2019)
3. Calder, M., Kolberg, M., Magill, E.H., Reiff-Marganiec, S.: Feature interaction: a critical review and considered forecast. Computer Networks **41**(1), 115–141 (2003)
4. Devriendt, F., Van Belle, J., Guns, T., Verbeke, W.: Learning to rank for uplift modeling. IEEE Transactions on Knowledge and Data Engineering **34**(10), 4888–4904 (2020)
5. Du, N., Dai, H., Trivedi, R., Upadhyay, U., Gomez-Rodriguez, M., Song, L.: Recurrent marked temporal point processes: Embedding event history to vector. In: Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining. pp. 1555–1564 (2016)
6. Gubela, R.M., Lessmann, S., Jaroszewicz, S.: Response transformation and profit decomposition for revenue uplift modeling. European Journal of Operational Research **283**(2), 647–661 (2020)
7. Guo, H., Tang, R., Ye, Y., Li, Z., He, X.: Deepfm: A factorization-machine based neural network for ctr prediction. In: Proceedings of the 26th International Joint Conference on Artificial Intelligence. pp. 1725–1731 (2017)
8. He, B., Weng, Y., Tang, X., Cui, Z., Sun, Z., Chen, L., He, X., Ma, C.: Rankability-enhanced revenue uplift modeling framework for online marketing. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. ACM (2024)
9. Kane, K., Lo, V.S., Zheng, J.: Mining for the truly responsive customers and prospects using true-lift modeling: Comparison of new and existing methods. Journal of Marketing Analytics **2**, 218–238 (2014)

10. Künzel, S.R., Sekhon, J.S., Bickel, P.J., Yu, B.: Metalearners for estimating heterogeneous treatment effects using machine learning. In: Proceedings of the national academy of sciences. vol. 116, pp. 4156–4165 (2019)
11. Lim, B.: Forecasting treatment responses over time using recurrent marginal structural networks. Advances in neural information processing systems **31** (2018)
12. Liu, D., Tang, X., Gao, H., Lyu, F., He, X.: Explicit feature interaction-aware uplift network for online marketing. In: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. ACM (2023)
13. Liu, R., Hou, Z.: Unite: A unified treatment effect estimation method for one-sided and two-sided marketing. In: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management. pp. 1472–1481 (2023)
14. Reutterer, T., Mild, A., Natter, M., Taudes, A.: A dynamic segmentation approach for targeting and customizing direct marketing campaigns. Journal of Interactive Marketing **20**(3-4), 43–57 (2006)
15. Rubin, D.B.: Causal inference using potential outcomes: Design, modeling, decisions. J. Amer. Statist. Assoc. **100**(469), 322–331 (2005)
16. Shalit, U., Johansson, F.D., Sontag, D.: Estimating individual treatment effect: generalization bounds and algorithms. In: International Conference on Machine Learning. pp. 3076–3085. PMLR (2017)
17. Sun, Z., Han, Q., Zhu, M., Gong, H., Liu, D., Ma, C.: Robust uplift modeling with large-scale contexts for real-time marketing. In: Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining. ACM (2025). https://doi.org/10.1145/3690624.3709293
18. Wang, X., Liu, T., Miao, J.: A deep probabilistic model for customer lifetime value prediction. arXiv preprint arXiv:1912.07753 (2019)
19. Wu, A., Kuang, K., Xiong, R., Li, B., Wu, F.: Stable estimation of heterogeneous treatment effects. In: International Conference on Machine Learning. pp. 37496–37510. PMLR (2023)
20. Yao, L., Li, S., Li, Y., Huai, M., Gao, J., Zhang, A.: Representation learning for treatment effect estimation from observational data. Advances in neural information processing systems **31** (2018)
21. Zhang, W., Li, J., Liu, L.: A unified survey of treatment effect heterogeneity modelling and uplift modelling. ACM Computing Surveys (CSUR) **54**(8), 1–36 (2021)
22. Zhang, X., Wang, K., Wang, Z., Du, B., Zhao, S., Wu, R., Shen, X., Lv, T., Fan, C.: Temporal uplift modeling for online marketing. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. ACM (2024)
23. Zhong, K., Xiao, F., Ren, Y., Liang, Y., Yao, W., Yang, X., Cen, L.: Descn: Deep entire space cross networks for individual treatment effect estimation. In: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. pp. 4612–4620 (2022)
24. Zhou, H., Huang, R., Li, S., Jiang, G., Zheng, J., Cheng, B., Lin, W.: Decision focused causal learning for direct counterfactual marketing optimization. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. ACM (2024)

A Theoretical Analysis of Ranking Error

In this section, we provide a theoretical justification for prioritizing a pairwise ranking objective over the standard Mean Squared Error (MSE). We demonstrate that minimizing MSE is a sufficient but not necessary condition for optimal

Algorithm 1 Implementation of ZILN-Euclidean Splitting Criterion

Require: Node samples (Y, T), Priors $(\bar{p}, \bar{\mu}, \bar{\sigma})$, Hyperparams (α_p, α_{reg})
Ensure: Gain $\Delta_{Gain}(s)$
Partition samples into Left (L) and Right (R) based on split s
▷ Strict Check: Min samples required in Treatment AND Control
if $\min(\sum T_L, \sum(1-T_L)) < 2$ **or** $\min(\sum T_R, \sum(1-T_R)) < 2$ **then return** 0
end if
function GetUplift(y, t)
  $p_T, \mu_T, \sigma_T \leftarrow$ CalcRobustParams$(y[t=1])$
  $p_C, \mu_C, \sigma_C \leftarrow$ CalcRobustParams$(y[t=0])$
  $E_T \leftarrow p_T \cdot \exp(\mu_T + 0.5\sigma_T^2)$
  $E_C \leftarrow p_C \cdot \exp(\mu_C + 0.5\sigma_C^2)$ **return** $E_T - E_C$
end function
function CalcRobustParams(y)
  $n \leftarrow |y|, \quad n_{pos} \leftarrow \sum \mathbb{I}(y > 0)$
  $\hat{p} \leftarrow (n_{pos} + \alpha_p \bar{p})/(n + \alpha_p)$
  if $n_{pos} > 1$ **then**
    $\mu_s, \sigma_s \leftarrow \text{moments}(\log(y[y > 0]))$
    $w \leftarrow n_{pos}/(n_{pos} + \alpha_{reg})$
    $\hat{\mu} \leftarrow w\mu_s + (1-w)\bar{\mu}$
    $\hat{\sigma} \leftarrow w\sigma_s + (1-w)\bar{\sigma}$
  else
    $\hat{\mu} \leftarrow \bar{\mu}, \quad \hat{\sigma} \leftarrow \bar{\sigma}$
  end if
  $\hat{\sigma} \leftarrow \text{clip}(\hat{\sigma}, 0.1, 4.0)$ **return** $\hat{p}, \hat{\mu}, \hat{\sigma}$
end function
$\tau_L \leftarrow$ GetUplift(Y_L, T_L)
$\tau_R \leftarrow$ GetUplift(Y_R, T_R)
▷ Maximize Euclidean distance weighted by sample size
$N_L \leftarrow |Y_L|, \quad N_R \leftarrow |Y_R|$
$Gain \leftarrow \frac{N_L \cdot N_R}{(N_L+N_R)^2} \cdot (\tau_L - \tau_R)^2$
return $Gain$

ranking, and frequently leads to suboptimal resource allocation in zero-inflated B2B distributions.

A.1 The Inadequacy of MSE for Ranking

Standard uplift methods (e.g., T-Learner) typically minimize the Pointwise Estimation Error (PEHE):

$$\epsilon_{PEHE} = \mathbb{E}_{x \sim \mathcal{X}}[(\hat{\tau}(x) - \tau(x))^2] \tag{14}$$

However, the business objective in sales optimization is to maximize the Qini coefficient (or AUUC), which depends solely on the *ordering* of individuals, not the precise magnitude of the prediction. A perfect ranking requires only that for any pair of individuals i, j:

$$\text{sign}(\hat{\tau}_i - \hat{\tau}_j) = \text{sign}(\tau_i - \tau_j) \tag{15}$$

MSE minimizes the squared magnitude difference. In B2B data where the true treatment effect $\tau(x) \approx 0$ for $> 80\%$ of samples (the "zero mass"), a model can achieve near-zero MSE by simply predicting $\hat{\tau}(x) = 0$ for the entire population. This "collapse to the mean" is catastrophic for ranking metrics, as it creates a random ordering among the zero-mass accounts and the low-lift positive accounts, yielding a Qini coefficient close to zero despite low MSE.

A.2 Optimality of Value-Weighted Ranking

We formally characterize the proposed Value-Weighted Ranking Loss ($\mathcal{L}_{V-Rank}$) as optimizing a lower bound on the revenue capture.

Proposition 1. *Minimizing the Value-Weighted Ranking Loss is equivalent to maximizing a smooth, convex lower bound on the Value-Weighted Pairwise Accuracy, thereby directly optimizing the expected revenue capture (Qini) rather than pointwise fit.*

Sketch of Proof. Let $\mathcal{D}$ be the dataset. The total ranking error (loss of potential revenue due to mis-ordering) can be defined as the sum of value differences for all discordantly ranked pairs:

$$\mathcal{L}_{TrueRank} = \sum_{i,j} \mathbb{I}(\text{sign}(\hat{\tau}_{ij}) \neq \text{sign}(\tau_{ij})) \cdot |\tau_i - \tau_j| \tag{16}$$

where $\tau_{ij} = \tau_i - \tau_j$ is the ground truth difference and $\hat{\tau}_{ij}$ is the predicted difference. This objective is discrete and non-differentiable.

Our proposed loss $\mathcal{L}_{V-Rank}$ introduces a soft surrogate using the logistic function $\phi(z) = \log(1 + e^{-z})$ and a weight $w_{ij} \approx \log(1 + |\tau_i - \tau_j|)$ derived from the transformed outcome Z:

$$\mathcal{L}_{V-Rank} \approx \sum_{i,j} w_{ij} \cdot \phi(\text{sign}(\tau_{ij}) \cdot \hat{\tau}_{ij}) \tag{17}$$

Since the logistic loss $\phi(z)$ is a convex upper bound on the 0-1 indicator loss, minimizing $\mathcal{L}_{V-Rank}$ minimizes the upper bound of the weighted ranking error. Unlike standard RankNet formulations (which set $w_{ij} = 1$), our weighting term w_{ij} ensures that gradients are dominated by pairs with large value discrepancies $|\tau_i - \tau_j|$. Consequently, the optimizer is tolerant of minor errors (e.g., mis-ranking two small accounts) but aggressively corrects "catastrophic inversions" (e.g., ranking a high-value account below a zero-value account), thereby directly maximizing the Area Under the Uplift Curve (AUUC).